
Report : timing

-path full

-delay max

-max_paths 5

Design : gf128_karatsuba_pipe_gf64

Version: V-2023.12-SP4

Date : Mon Feb 16 22:03:54 2026

Operating Conditions: ff0p715v125c Library: saed14rvt_base_ff0p715v125c

Wire Load Model Mode: top

Startpoint: b[13] (input port clocked by clk)

Endpoint: mul_p1/stage1_sum_reg[40]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Des/Clust/Port Wire Load Model Library

gf128_karatsuba_pipe_gf64

8000

saed14rvt_base_ff0p715v125c

Point	Incr	Path
-----	-----	-----
clock clk (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
input external delay	0.50	0.50 f
b[13] (in)	0.00	0.50 f
U348/X (SAEDRVT14_E02_0P5)	0.34	0.84 r
mul_p1/b[13] (gf64_mul_pipe2_1)	0.00	0.84 r
mul_p1/U106/X (SAEDRVT14_INV_S_2)	0.47	1.31 f
mul_p1/U1045/X (SAEDRVT14_INV_S_2)	0.51	1.82 r
mul_p1/U5966/X (SAEDRVT14_ND2_CDC_1)	0.09	1.91 f
mul_p1/U5967/X (SAEDRVT14_EN4_M_1)	0.07	1.98 f
mul_p1/U614/X (SAEDRVT14_EN4_M_1)	0.07	2.04 f
mul_p1/U505/X (SAEDRVT14_EN4_M_1)	0.08	2.13 r
mul_p1/U244/X (SAEDRVT14_OAI21_0P5)	0.04	2.16 f
mul_p1/U361/X (SAEDRVT14_AOI21_0P5)	0.06	2.23 r
mul_p1/stage1_sum_reg[40]/D (SAEDRVT14_FDPRBQ_V2LP_0P5)	0.00	2.23 r
data arrival time		2.23
clock clk (rise edge)	2.50	2.50
clock network delay (ideal)	0.00	2.50
clock uncertainty	-0.20	2.30
mul_p1/stage1_sum_reg[40]/CK (SAEDRVT14_FDPRBQ_V2LP_0P5)	0.00	2.30 r
library setup time	-0.01	2.29
data required time		2.29
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data required time		2.29
data arrival time		-2.23
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slack (MET)		0.06

Startpoint: b[13] (input port clocked by clk)

Endpoint: mul_p1/stage1_sum_reg[50]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Des/Clust/Port Wire Load Model Library

 gf128_karatsuba_pipe_gf64
 8000

saed14rvt_base_ff0p715v125c

Point	Incr	Path
-----	-----	-----
clock clk (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
input external delay	0.50	0.50 f
b[13] (in)	0.00	0.50 f
U348/X (SAEDRVT14_E02_0P5)	0.34	0.84 r
mul_p1/b[13] (gf64_mul_pipe2_1)	0.00	0.84 r
mul_p1/U106/X (SAEDRVT14_INV_S_2)	0.47	1.31 f
mul_p1/U1045/X (SAEDRVT14_INV_S_2)	0.51	1.82 r
mul_p1/U1043/X (SAEDRVT14_ND2_CDC_1)	0.08	1.89 f
mul_p1/U833/X (SAEDRVT14_EN4_M_1)	0.07	1.97 f
mul_p1/U639/X (SAEDRVT14_EN4_M_1)	0.07	2.03 f
mul_p1/U510/X (SAEDRVT14_EN4_M_1)	0.08	2.11 r
mul_p1/U236/X (SAEDRVT14_OAI21_0P5)	0.04	2.15 f
mul_p1/U326/X (SAEDRVT14_AOI21_0P5)	0.06	2.22 r
mul_p1/stage1_sum_reg[50]/D (SAEDRVT14_FDPRBQ_V2_0P5)	0.00	2.22 r
data arrival time		2.22
clock clk (rise edge)	2.50	2.50
clock network delay (ideal)	0.00	2.50
clock uncertainty	-0.20	2.30
mul_p1/stage1_sum_reg[50]/CK (SAEDRVT14_FDPRBQ_V2_0P5)	0.00	2.30 r
library setup time	-0.02	2.28
data required time		2.28
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data required time		2.28
data arrival time		-2.22
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slack (MET)		0.06

Startpoint: b[13] (input port clocked by clk)

Endpoint: mul_p1/stage1_sum_reg[52]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Des/Clust/Port Wire Load Model Library

 gf128_karatsuba_pipe_gf64
 8000

saed14rvt_base_ff0p715v125c

Point	Incr	Path
-----	-----	-----
clock clk (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
input external delay	0.50	0.50 f
b[13] (in)	0.00	0.50 f
U348/X (SAEDRVT14_E02_0P5)	0.34	0.84 r
mul_p1/b[13] (gf64_mul_pipe2_1)	0.00	0.84 r
mul_p1/U106/X (SAEDRVT14_INV_S_2)	0.47	1.31 f
mul_p1/U1045/X (SAEDRVT14_INV_S_2)	0.51	1.82 r
mul_p1/U5928/X (SAEDRVT14_ND2_CDC_1)	0.08	1.90 f
mul_p1/U5930/X (SAEDRVT14_E04_1)	0.07	1.97 r
mul_p1/U616/X (SAEDRVT14_EN4_M_1)	0.06	2.03 f
mul_p1/U499/X (SAEDRVT14_EN4_M_1)	0.08	2.11 r
mul_p1/U240/X (SAEDRVT14_OAI21_0P5)	0.04	2.15 f
mul_p1/U396/X (SAEDRVT14_AOI21_0P5)	0.06	2.22 r
mul_p1/stage1_sum_reg[52]/D (SAEDRVT14_FDPRBQ_V2_0P5)		

	0.00	2.22 r
data arrival time		2.22
clock clk (rise edge)	2.50	2.50
clock network delay (ideal)	0.00	2.50
clock uncertainty	-0.20	2.30
mul_p1/stage1_sum_reg[52]/CK (SAEDRVT14_FDPRBQ_V2_0P5)	0.00	2.30 r
library setup time	-0.02	2.28
data required time		2.28

data required time		2.28
data arrival time		-2.22

slack (MET)		0.06

Startpoint: b[13] (input port clocked by clk)
 Endpoint: mul_p1/stage1_sum_reg[56]
 (rising edge-triggered flip-flop clocked by clk)
 Path Group: clk
 Path Type: max

Des/Clust/Port	Wire Load Model	Library
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gf128_karatsuba_pipe_gf64		
8000		saed14rvt_base_ff0p715v125c

Point	Incr	Path

clock clk (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
input external delay	0.50	0.50 f
b[13] (in)	0.00	0.50 f
U348/X (SAEDRVT14_E02_0P5)	0.34	0.84 r
mul_p1/b[13] (gf64_mul_pipe2_1)	0.00	0.84 r
mul_p1/U106/X (SAEDRVT14_INV_S_2)	0.47	1.31 f
mul_p1/U1045/X (SAEDRVT14_INV_S_2)	0.51	1.82 r
mul_p1/U1042/X (SAEDRVT14_ND2_CDC_1)	0.08	1.90 f
mul_p1/U854/X (SAEDRVT14_E04_1)	0.07	1.97 f
mul_p1/U546/X (SAEDRVT14_EN4_M_1)	0.06	2.03 f
mul_p1/U457/X (SAEDRVT14_EN4_M_1)	0.08	2.11 r
mul_p1/U4985/X (SAEDRVT14_OAI21_0P5)	0.04	2.15 f
mul_p1/U330/X (SAEDRVT14_AOI21_0P5)	0.06	2.22 r
mul_p1/stage1_sum_reg[56]/D (SAEDRVT14_FDPRBQ_V2_0P5)	0.00	2.22 r
data arrival time		2.22
clock clk (rise edge)	2.50	2.50
clock network delay (ideal)	0.00	2.50
clock uncertainty	-0.20	2.30
mul_p1/stage1_sum_reg[56]/CK (SAEDRVT14_FDPRBQ_V2_0P5)	0.00	2.30 r
library setup time	-0.02	2.28
data required time		2.28

data required time		2.28
data arrival time		-2.22

slack (MET)		0.06

Startpoint: b[13] (input port clocked by clk)
 Endpoint: mul_p1/stage1_sum_reg[34]
 (rising edge-triggered flip-flop clocked by clk)
 Path Group: clk

Path Type: max

Des/Clust/Port Wire Load Model Library

gfl28_karatsuba_pipe_gf64
8000

saed14rvt_base_ff0p715v125c

Point	Incr	Path

clock clk (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
input external delay	0.50	0.50 f
b[13] (in)	0.00	0.50 f
U348/X (SAEDRVT14_E02_0P5)	0.34	0.84 r
mul_p1/b[13] (gf64_mul_pipe2_1)	0.00	0.84 r
mul_p1/U106/X (SAEDRVT14_INV_S_2)	0.47	1.31 f
mul_p1/U2117/X (SAEDRVT14_INV_S_1)	0.50	1.80 r
mul_p1/U4765/X (SAEDRVT14_ND2_CDC_1)	0.09	1.89 f
mul_p1/U871/X (SAEDRVT14_E04_1)	0.07	1.96 f
mul_p1/U589/X (SAEDRVT14_EN4_M_1)	0.06	2.03 f
mul_p1/U538/X (SAEDRVT14_EN4_M_1)	0.08	2.11 r
mul_p1/U1377/X (SAEDRVT14_OAI21_0P5)	0.04	2.15 f
mul_p1/U370/X (SAEDRVT14_AOI21_0P5)	0.06	2.21 r
mul_p1/stage1_sum_reg[34]/D (SAEDRVT14_FDPRBQ_V2_0P5)	0.00	2.21 r
data arrival time		2.21
clock clk (rise edge)	2.50	2.50
clock network delay (ideal)	0.00	2.50
clock uncertainty	-0.20	2.30
mul_p1/stage1_sum_reg[34]/CK (SAEDRVT14_FDPRBQ_V2_0P5)	0.00	2.30 r
library setup time	-0.02	2.28
data required time		2.28

data required time		2.28
data arrival time		-2.21

slack (MET)		0.07